Business Insights Report

Bank Marketing Campaign & Customer Response Analytics ·

1. Purpose and data

This report summarises what the data show about which bank customers subscribe to the term deposit and how the telephone campaign behaved. It covers 11,162 customers. In Sprint 1 the raw file was cleaned entirely in Python: disguised missing-value text ('unknown', 'not_available', 'missing') was converted to real missing values; 12 impossible ages (122–146) and 15 balances, 12 call durations and 10 contact counts far outside the original release's range were treated as data-entry errors; remaining gaps were filled with medians (numbers) or the existing 'unknown' category (labels). No duplicates were found and no rows were deleted.

All findings below are **associations**, not proven causes. Statistical validation is in the Statistical Analysis Report, and actions are in the Final Recommendations.

2. Key numbers at a glance

Measure	Value
Overall response rate	47.4% (5,289 of 11,162 subscribed)
Typical customer	age 39, account balance 551, call of about 4 minutes, 2 contacts
Best-responding groups	debt-free customers aged 60+ (79.4%) or under 30 (75.9%)
Weakest large group	blue-collar customers with a housing loan (32.7%, 1,306 customers)
Previous campaign success	91.3% subscribe again
Unknown contact method	only 23.8% subscribe

3. Who the customers are

- Jobs: management (22.8%), blue-collar (17.2%) and technician (16.2%) make up over half the base.

- 56.9% are married; secondary education is the most common (48.7%), then tertiary (32.7%).
- Contact: cellular 71.4%, unknown 21.7%, telephone 6.9%. 74.9% had no earlier campaign.
- Finances: 47.3% have a housing loan, 13.1% a personal loan, 1.5% are in default.
- Balance is highly skewed: the median is 551 but the mean is 1,520, so the median is the honest 'typical' figure.
- Campaign effort was uneven: May alone holds 2,824 contacts (25% of the campaign).

4. Who responds

4.1 Job, age and demographics

- Students (74.7%, 356 customers) and retirees (66.4%, 771) respond most; unemployed 56.6%. Blue-collar (36.1%) and entrepreneurs (37.6%) respond least.
- Age is U-shaped: under 30 → 59.7%, 30–39 → 44.4%, 40–49 → 40.5%, 50–59 → 41.8%, 60+ → 77.0%. Students and retirees drive the two ends.
- Tertiary education 54.2% vs primary 39.2%; single 54.3% vs married 43.4%. Job, education and marital status overlap, so these effects are not independent.

4.2 Financial position and loans

- Higher balance, higher response: overdrawn 30.5%, low (0–500) 42.5%, medium (500–2,000) 49.9%, high (2,000+) 57.8%.
- Housing loan 36.6% vs 57.0% without; personal loan 33.2% vs 49.5% without.
- By debt status: no loans 59.6%, housing only 37.6%, personal only 35.0%, both 31.7%.
- Subscribers hold a higher median balance in every job group.

4.3 Customer segments (age × debt status, groups of 100+ customers)

Segment	Customers	Response rate
Debt-free, age 60+	660	79.4%
Debt-free, under 30	760	75.9%
Age 50–59, housing loan only	585	29.7%
Age 50–59, housing + personal loan	126	28.6%

Every debt-free age group beats the 47.4% baseline; every group with a loan falls below it.

5. How the campaign behaved

- **Contact method:** cellular 54.2%, telephone 50.5%, unknown 23.8%. Of 2,423 unknown-contact customers, 76.2% said no. 92% of these records fall in May and June, so this may be a record-keeping or lead-source issue rather than a true channel effect.
- **Previous campaign:** previous success 91.3% (971 of 1,063), other 57.0%, failure 49.9%, no history 40.8%. Customers contacted before respond at 67.1% vs 40.7% for new customers. Number of earlier contacts barely matters (66–69%).
- **Number of contacts:** 1 call 53.4%, 2 calls 46.3%, 3 calls 46.7%, 4–5 calls 39.7%, 6+ calls 29.3%. Customers who keep refusing are the ones called again, so this shows an association, not that extra calls cause refusals.
- **Call duration:** under 2 minutes 9.3%, 2–5 minutes 39.8%, 5–10 minutes 61.7%, over 10 minutes 85.1%. Subscribers' median call is 418 s vs 165 s. Duration is known only after the call, so it measures call quality, not whom to call.
- **Timing:** response is highest in low-volume months (March 89.9%, September 84.3%, October 82.4%, December 90.9%) and lowest in the busiest, May (32.8%). Month, volume and contact method are tangled together, so this cannot be read as a pure month effect.
- **Correlations with subscribing:** duration +0.45 (the only moderate one), pdays +0.15, previous contacts +0.14, number of contacts −0.13, balance +0.08, age +0.04.

6. What the campaign costs

6.1 Customer-level cost (assumption-based proxy)

The dataset has no cost or revenue fields, so a proxy was built from two **assumed** constants: 0.50 per call-minute and 200 per subscriber. These are illustrative, not bank financials; the 2,970% overall 'ROI' they produce is a product of those assumptions and must not be quoted as a real return.

- 31.7% of call-time cost is spent on customers who said no.
- Average cost per subscriber is 6.52. It is highest for unknown-contact customers (12.74 vs about 5.75), 6+ calls (9.17 vs 5.61 for one call) and blue-collar customers (9.05); lowest for previous successes (3.16 vs 7.75 for no history), students (3.68) and retirees (4.86).

6.2 Real marketing campaign table (88 campaigns, analysed separately)

This table has no customer key and uses different channel names, so it is not joined to customers.

- Pooled ROI by channel: Social Media 379%, PPC 353%, Email 313%, Referral 295%, Paid Search 228% (5 campaigns with no channel: 161%). 7 of 88 campaigns lost money.
- Spend and revenue are almost unrelated (correlation +0.06); clicks and revenue −0.03. Email earns the least revenue per click (7.57 vs about 9.7–10.5).
- With only 11–22 campaigns per channel, these channel differences were not statistically significant (see the Statistical Analysis Report).

7. What these findings do not prove

- Associations only: capping calls or changing channel has not been tested as an intervention.
- Some groups are small (default = yes 168 customers, December 110, unknown job about 170).
- ROI figures rest on assumed constants; the campaign table is small and unlinked to customers.
- Duration cannot be used for pre-call targeting.